

UMF — Covering Note

Provenance, deliverable, and reading order for the Unified Metric Framework documentation
Stephen L Browne | Central Coast NSW, May 2026 (AEST) | v1.3

Thank you for taking the time to read this work. The package is publicly available at <https://github.com/OldGreyWhistle/The-UMF> and consists of four core documents plus one companion piece, totalling about forty pages. This covering note frames the rest. The other documents are:

- **Foundational Logic v1.3.** The five-postulate structural commitments underlying the framework. Read this first.
- **Formal Framework v2.3.** The current commitments at the operational level — T_G as the load-bearing dynamical field, the wavefront's totally reflective inner boundary, the variable electromagnetic vacuum couplings, the dark matter formalism, the stochastic gravitational-wave background internal to Region A, the blast-metrics regime as the analogical framing for the establishing event, the standing rules, and the open theoretical questions. Read this second; it is the longest of the documents and is where the substantive theoretical content lives.
- **Empirical Engagement v1.3.** The observational programme and what the framework does and does not currently engage. Read this third.
- **Bounded-Universe Topology v1.0 (companion).** A literature-review-and-positioning analysis of the bounded-universe topology question under the reflective-boundary commitment. The first UMF-specific empirical engagement undertaken under the standing rules without specialist input. Read this fourth, after the core documents. Surveys matched-circles methodology (Cornish-Spergel-Starkman 1998 onward), Planck topology bounds, Israel junction condition apparatus (Israel 1966, Berezin-Kuzmin-Tkachev 1987, contemporary applications), and identifies five specific specialist work items.

Three cold reviews have been performed on the v2.0 package: by the research lead (the author), by ChatGPT, and by Claude Opus 4.7 in incognito mode. The v2.1 documents incorporated the substantive feedback from those reviews. The v2.2 documents incorporated further clarifications of existing framework commitments (the wavefront's totally reflective inner boundary, the stochastic gravitational-wave background framing, the blast-metrics regime naming, and reference to the framework lineage's prior empirical work under UMDT). The v2.3 documents (current) fold in housekeeping refinements consequent on public GitHub publication, the published bounded-universe topology positioning document, additional open questions raised by post-V2.2 commentary (entropy at the reflective boundary, DM scaling with cosmic time, potential simultaneous resolution of S_8 / BBN / CMB acoustic peaks tensions through DM behaviour), and the specific ISW meta-analysis constraint in the framework lineage section.

The cold review reports and the V2.2 commentary are available on request if useful context. The GitHub repository contains all versions for revision history.

What I am asking for

I am asking for a triage-level evaluation of the framework rather than for theoretical derivation. Specifically, I am asking you to read the package end-to-end and write a response of three to five pages addressing four questions.

- **Are the foundational logic and the framework's structural commitments physically defensible?** A specialist who finds them unsound can stop there; the formal scaffold is not worth evaluating if the foundations are rejected. A specialist who finds them at least plausible should proceed to the remaining questions.
- **Of the open theoretical questions listed in the Formal Framework document, which is most tractable as a standalone problem?** Both prior cold reviews independently recommended the variable-coupling Maxwell sector using Bekenstein/

BSBM formalism, with the action principle flagged as the gating issue. I would value your independent judgement on this, including whether you concur with the prior reviews or see a different priority. What additional commitments would the framework need to make for that question to be addressable, and what would a quantitative attempt at the question look like in outline?

- **Are there showstoppers the framework has not flagged?** Internal inconsistencies, conflicts with established empirical results, or structural problems that the prior cold reviews missed and that I have not flagged. The v2.3 documents flag the action principle issue, the Birkhoff theorem issue (partially defused by the stochastic SGWB reframing), the CMB acoustic peaks issue, the inflation/horizon problem issue, the wave-energy distinction in the blast-metrics regime, the relaxation-producing- $w \approx -1$ mechanism gap, entropy at the reflective boundary, DM scaling with cosmic time, and the potential simultaneous resolution of S_8 / BBN / CMB acoustic peaks tensions through DM behaviour. I want to know what remains undetected at this stage of development.
- **Where does the framework most need formal apparatus from existing literature?** Both prior reviews identified scalar-tensor theory (Brans-Dicke, Horndeski), Bekenstein/BSBM varying-constants formalism, Israel junction conditions for the wavefront, modified-gravity DM theories (TeVes, MOG, Verlinde), and holographic boundary formalisms. The bounded-universe topology positioning document (v1.0) confirms Israel junction conditions as the relevant apparatus for the wavefront, citing contemporary peer-reviewed applications (Schubert 2026 in Entropy). I would value your judgement on which of these is most aligned with the framework's commitments and what specific work would be needed to import that apparatus under the framework's evaluation rule.

I am not asking for derivations, calculations, or theory generation. I am asking for evaluation and triage. This is achievable in two to three days of careful reading and writing and is what I am paying \$2000 for. If the work suggests that more is needed, we can talk about that separately; the initial engagement is bounded at this scope.

Provenance

The structural intuitions in this framework — gravitational primacy, a bounded cosmological-scale gravitational event with an actively reflective wavefront, relaxation as the source of dark-energy phenomenology, finite cosmic lifecycle — have been my working position since 1983, formed during a period when both black holes as physical objects and gravitational waves as observables were contested or unconsidered in the rooms I was in. The formal scaffold is recent; the structural picture is not.

I was a B Eng and BSc student at the University of Sydney in 1983, working on the TORTUS tokamak in plasma physics. I was looking at superconductivity for my 1984 thesis. I was hospitalised in November 1983 and missed my final exams. I never returned to formal physics. I pivoted into electrical engineering, joining Ferranti Computer Systems Australia, where I worked through the 1980s and into the 1990s. After Ferranti PLC was undermined by the International Signal and Control fraud, I was hardware team leader from tender phase onward on the NSW State Rail Authority Train Describer and Telemetry System. I retired from professional engineering in recent years. I live on the Central Coast of NSW (Australian Eastern Standard Time, applicable in standard-time months April through October; Australian Eastern Daylight Time applies in daylight-time months October through April).

The cosmological work has been a parallel personal interest across that career, sustained by reading and thinking rather than by formal training. The structural intuitions have remained stable across four decades. The recent formalisation work

— through several revisions across late 2025 and early 2026 — has used LLM-loop iteration (primarily with Claude, with some cross-checking through ChatGPT and Gemini) to formalise the structural intuitions into a written framework. I am aware of the limitations of LLM-loop methodology and have caught several regressions where the iterative work drifted away from my actual structural commitments. The framework as it stands is the product of structural intuitions I have held for forty-three years, formalised through methodology I am explicitly aware of the limits of.

I am not an academic and I make no credential claim. The framework should be evaluated on its merits, with the provenance context informing the reading rather than supporting it. The engineering provenance (Ferranti, SRA Train Describer) is relevant because the methodological discipline I bring to the framework — rigorous verification, refusal to let plausibility substitute for derivation, careful separation of what is committed from what is hand-waved, acceptance of brutal review and incorporation of it — is the discipline developed across that engineering career. The framework is the product of engineering methodology applied to a structural picture of cosmology, by someone who would have been a physicist in a different life but was not.

Methodological discipline

The framework has been developed under three standing rules that I want to flag explicitly, because they are integral to the framework's current state and the response document should be aware of them.

- **Rigour standard.** 5σ for proofs, 4σ for correlations, 2σ - 3σ notable only. The framework reports its empirical engagement at this level and does not claim support for itself from findings below 4σ . Findings at 2σ - 3σ are noted as "not in tension with" rather than as "alignment with" — a distinction that matters for the DR2 interpretation specifically.
- **Honesty in evaluation.** The framework presents its negative results alongside its positive ones, flags its speculative content as speculative, and distinguishes candidate commitments from adopted ones. The framework does not present itself as more developed than it is.
- **Full evaluation in UMF context before any import.** Any formalism, metric, or theoretical apparatus imported into the framework must be evaluated in full — units, regime of validity, mapping to UMF, consistency with other commitments — before it is adopted. The framework distinguishes adopted commitments from candidate imports that have not yet been evaluated.

The framework is also committed to permanent rejection of variable c . This was a regression in the V1.5 version that I caught on re-read; the commitment to invariant c is now recorded permanently because variable c breaks spacetime structure by definition. This is mentioned because it is the kind of methodological commitment that protects the framework against the failure modes of LLM-loop iteration, and it is the kind of commitment your evaluation should be aware of.

The framework lineage has conducted five pre-registered empirical analyses under the precursor framework UMDT (Pantheon+ residual spatial analysis, ISW cross-correlation meta-analysis, CMB echo parametric pumping, CMB lattice Path 1, and DESI DR2 BAO), with four nulls and one positive consistency match. One finding (DESI DR2 P2 phantom-crossing) reached 4σ correlation-class significance under UMDT's specific commitments. The ISW meta-analysis converted to a constraint of $|\alpha| < 0.3$ at 3σ on linear redshift trend in A_{ISW} ; the Pantheon+ analysis converted to a 1.1% upper limit on coherent fractional metric variation above 5.8×10^{-18} Hz. The empirical record demonstrates capacity for pre-registered work at the methodological standard the field expects. The bounded-universe topology positioning document (v1.0) is the first UMF-specific empirical engagement, at the structural-inference level. The full Empirical Engagement document expands on this lineage

and on how UMF-specific quantitative empirical engagement awaits the resolution of the framework's open theoretical questions.

What this work is and is not

The framework is a pre-paradigmatic structural picture, not a developed theory. It does not currently make quantitative predictions distinct from Λ CDM at observational precision. The framework is not in tension with DESI DR2 at notable significance (2σ - 3σ) but cannot claim alignment with DR2 at the framework's own rigour standard until DR3 provides correlation-class evidence (or fails to). The framework is honest about this status; it presents itself as a structural picture with the open theoretical questions flagged for specialist input, not as a developed theory ready for empirical engagement.

The framework does not claim to solve the major open problems in cosmology. It does not solve the H_0 tension (the V1.5 attempt produced a parameter-regime bound violation against BBN and LLR; the framework now acknowledges this). It does not produce the CMB power spectrum from first principles. It does not derive galactic rotation curves at quantitative level. It does not currently address the CMB acoustic peaks issue (the framework's gravitoelectric DM presupposes structures that do not exist at recombination); existing UMDT-era CMB analysis work may be relevant and is to be incorporated in a future revision. The framework does not provide a mechanism for inflation, the horizon problem, or the primordial power spectrum. These are work items for specialist development, not current capabilities.

What the framework does is offer a coherent alternative structural picture of the cosmological lifecycle, with specific commitments about the universe's bounded structure, its totally reflective active wavefront boundary, its relaxation-driven dark-energy phenomenology, and its finite lifecycle. The structural picture is internally consistent (with the consistency questions from earlier revisions either resolved or flagged), connects to existing physics formalism through candidate imports that need evaluation, and identifies specific empirical engagement opportunities. The first UMF-specific empirical engagement at the structural-inference level (bounded-universe topology under reflective-boundary commitment) is filed in the companion document v1.0.

My honest assessment is that the framework is at the stage of structural maturity where specialist input is necessary for further development. The LLM-loop methodology has reached its useful endpoint, with three independent cold reviews now in. The next stage of development requires either commitment to specific mathematical formalisms (which I cannot evaluate at the required level of rigour) or rejection of the framework on grounds the prior reviewers and I have not anticipated. Either outcome is acceptable to me; the framework should be evaluated on its merits and the evaluation should be honest. I have asked for that honesty explicitly in setting up this engagement.

Closing

Thank you again for taking the time to read this. The framework represents four decades of structural intuition and several months of formal scaffolding work. Whatever you make of it, your evaluation will be the most substantive engagement the framework has received from someone with formal training in the relevant physics. I am grateful for that, and I want the evaluation to be useful to you as well as to me — feel free to engage at whatever level interests you, and to be direct about both what works and what does not.

I have asked for brutal honesty throughout this work. The same applies to your response. I would rather hear that the framework has a fatal flaw the prior reviewers and I have not seen than have you soften the evaluation for my benefit. The methodological discipline I have brought to the work, and the willingness to absorb the three prior cold reviews and revise accordingly, is precisely the discipline that can absorb that kind of feedback without damage to the work or to the relationship.

I look forward to your response. Take whatever time you need within the bounded engagement.

With thanks,

Stephen L Browne

Central Coast NSW, May 2026 (AEST)

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